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Regulation: 2022

Reg. No.:

BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM – 638 401

PERIODICAL TEST I - JANUARY 2023

I Semester '

Stay Ahead

22CH103 & ENGINEERING CHEMISTRY I

Degree & Branch: B.E. / B.Tech. & Common to all branches

Time: 1:30 Hrs. Maximum: 50 Marks

Instructions:

- Students should not mark/write anything on the Question Paper other than the register number.
- Section A and B of the Question Paper contains questions for 30 Marks each. Section C contains questions for 15 Marks.
- Students can attempt answering any two out of three subsections in each section. Maximum mark is limited to 20 in section A & B, and 10 in sections C.

Q.No.	Questions		
	SECTION A	COURSE OUTCOME 1	MAXIMUM: 20 MARKS

A1

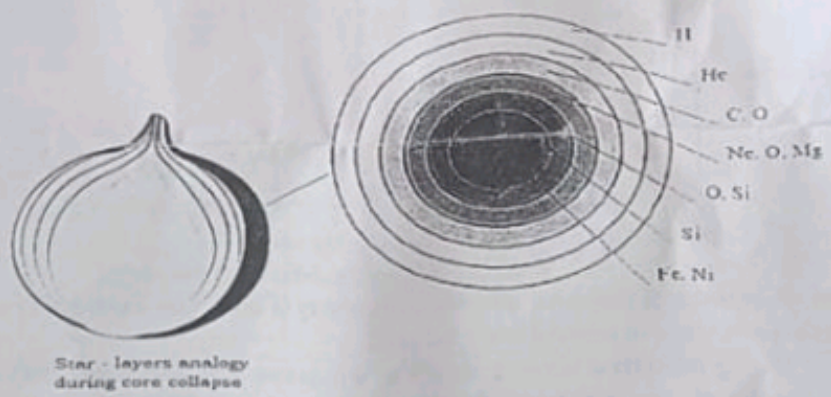
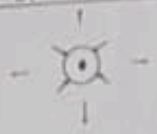


Figure 1. Schematic structure of fully evolved super giant star

The prominent burning stages analogous to onion-skin make-up is shown, which results in various elements that acts as a fuel for next stage.

(i)	Figure 1 depicts that the reaction starts with hydrogen burning leading to He→ C →O→ Ne→Mg→Si→Fe/Ni. Is there a probability to form silicon directly by hydrogen burning? Justify your answer. (3 marks-[U/C],2)
(ii)	Assume that the star depicted in figure 1 undergoes supernovae explosion at the end of the formation of iron/nickel. Identify the case (1 or 2) that will result in supernovae and justify your option. Case 1 A star like sun forming clouded dissipation of matter Case 2: Sudden collapse of a massive star forming elements up to Uranium (3 marks-[U/C],2)
(iii)	As the stellar core become dominated by iron, the different stages causing super novae is indicated below (Table 1]), arrange in sequence keeping the scenario in figure 1 as stage (a)

Table 1

(b) The inner part of the core is compressed into neutrons	(d) Energy absorbed by iron is greater than the energy released for fusion
	
(c) The surrounding material is blasted away, leaving only a degenerated remnant.	(e) In-falling material bounces back and forms outward propagating shock front

(4 marks-[An/C],2)

A2

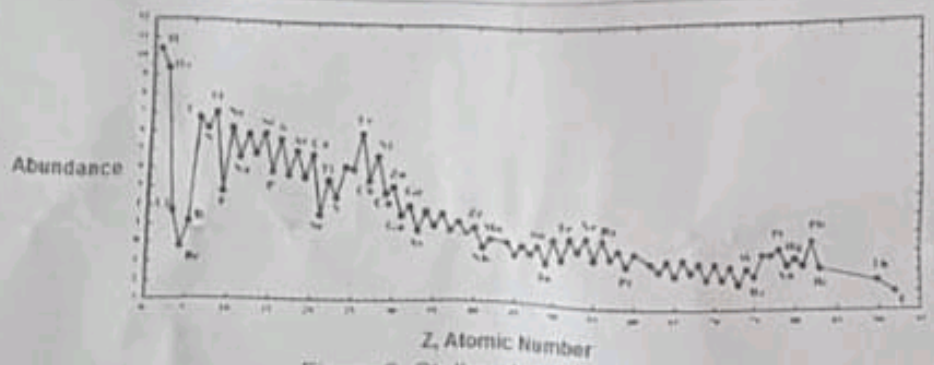


Figure 2 Stellar abundance of elements

The abundance of elements and nuclear stability is based on whether a nucleus contains an odd/even number of protons and neutrons.

(i) During advanced burning stages of a massive star with more than 8 solar masses, if alpha particle is reactant A, match the possible stable products P when B undergoes fusion with A given in the Table 2.

Table 2

S.No	Reactant (A)	Reactant (B)	Stable product (P)
1	^{4}He	^{12}C	$^{32}S + \gamma$
2	^{4}He	^{20}Ne	$^{36}Ar + \gamma$
3	^{4}He	^{28}Si	$^{16}O + \gamma$
4	^{4}He	^{32}S	$^{40}Ca + \gamma$
5	^{4}He	^{36}Ar	$^{24}Mg + \gamma$

(3 marks-[U/C],2)

(ii) Select the two processes (a-e from the list) that provide neutrons for s- process during the formation of elements heavier than iron in figure 2.

(a) $^{13}C + {}^{4}He \longrightarrow {}^{16}O + n$
(b) $^{13}Si + {}^{4}He \longrightarrow {}^{16}O + \gamma$
(c) $^{13}Ar + {}^{4}He \longrightarrow {}^{16}O + p$
(d) $^{13}S + {}^{4}He \longrightarrow {}^{16}O + e^{-}$
(e) $^{22}Ne + {}^{4}He \longrightarrow {}^{25}Mg + n$

(3 marks-[An/C],2)

(iii)

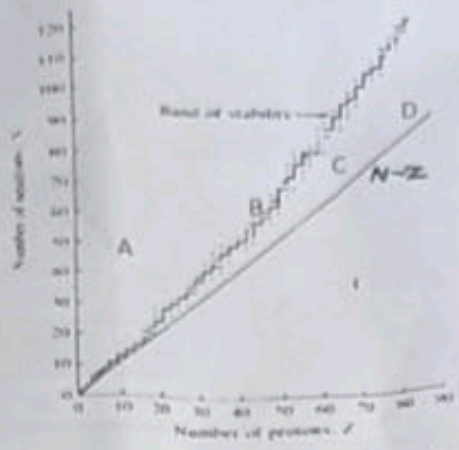


Figure 3 Graphical representation of band of stability

Identify the most stable nuclide among the following: Cl-34 or Cl-35 or Cl-36. Predict and map the given isotopes of Cl in the regions A/B/C/D (figure 3) based on abundance (figure 2) Hint: Atomic number of chlorine is 17.

(4 marks-[Ap/C],2)

A3

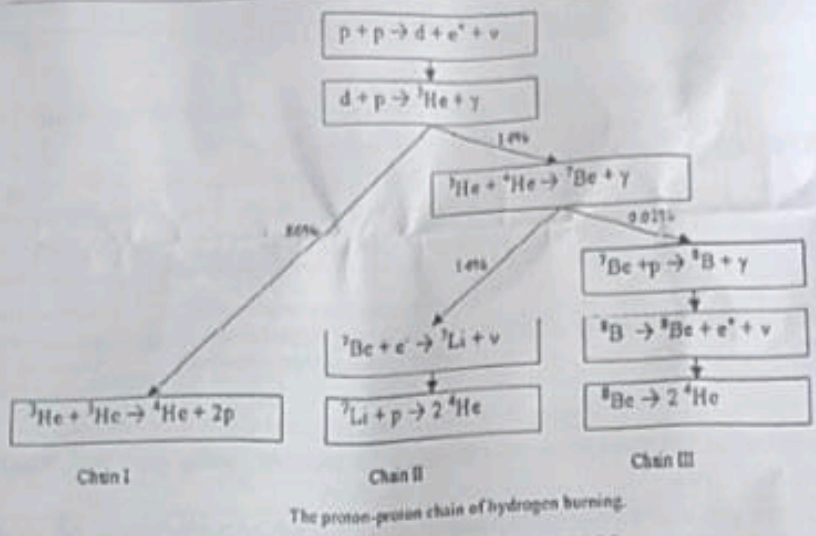


Figure 4 Hydrogen burning process

A fusion reaction taking place in a star is depicted in the image where protium combination leads to the formation of elements and release of energy.

(i) The process indicated (fig 4) takes place in a small star like sun. If the mass is greater than 1.3 solar masses, will the same process take place? Justify your answer.

(3 marks-[U/C],2)

(ii) Deduce the stable elements formed from the p-p chain other than hydrogen from figure 4a.

(3 marks-[Ap/C],2)

(iii)

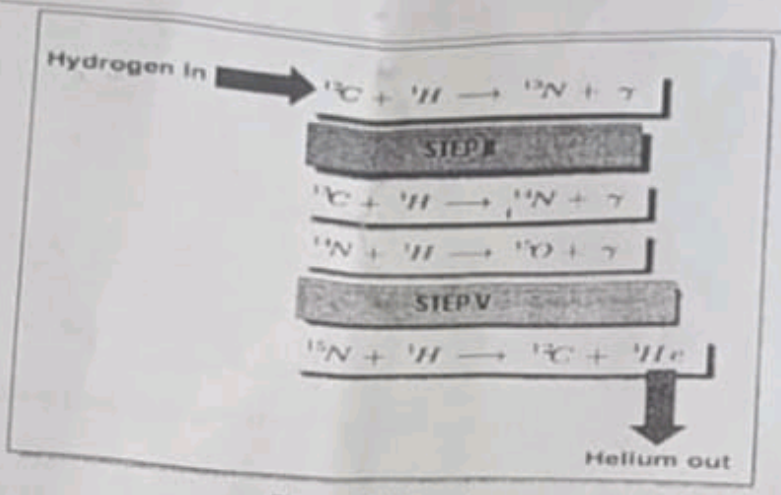


Figure 5 CNO cycle

Complete Step II and V of CNO cycle in figure 5 by referring to Figure 4b.

(4 marks-[An/C],2)

SECTION B

B1	COURSE OUTCOME 2	MAXIMUM : 20 MARKS

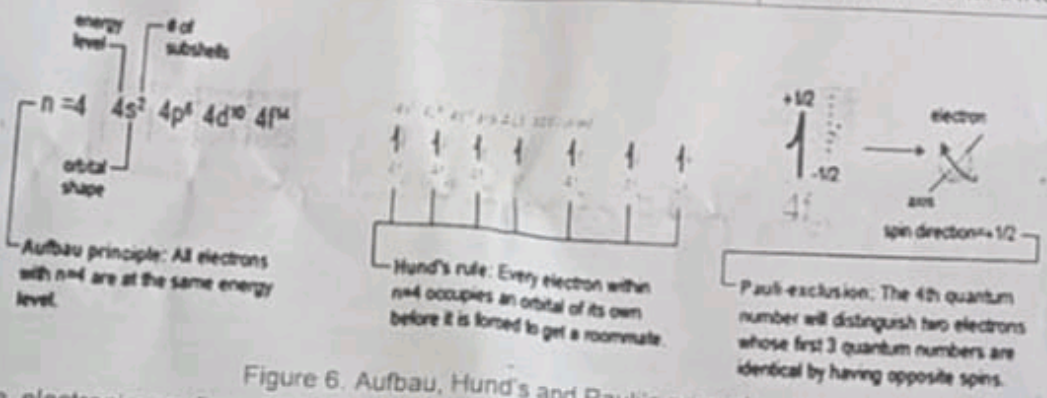


Figure 6. Aufbau, Hund's and Pauli's principles

The electronic configuration of elements in periodic table can be assigned based on the above set of rules.

(i) Write the electronic configuration of $_{42}Mo$ and $_{47}Ag$ elements. Apply Aufbau principle and predict it's suitability for stable electronic arrangements. [Refer fig 6]

(3 marks- [Ap/C],3)

(ii) Identify the set of possible four quantum numbers for the 7th electron of the 3d orbital ($3d^7$). [Refer figure 6]

(3 marks- [U/C],2)

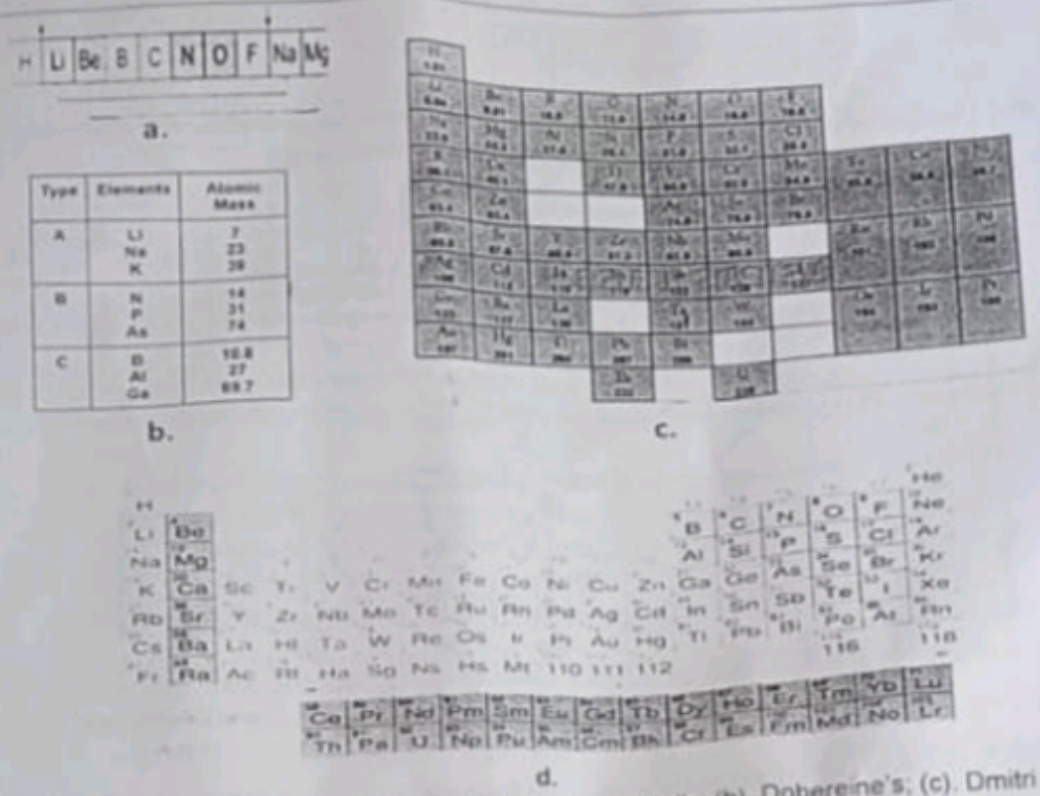
(iii) Complete the following table 3. [Refer figure 6]

Table 3

Atomic number	Mass number	Number of neutrons	Number of Protons	Number of Electrons	Name of the atomic species
-	1	0	1	0	-
5	-	5	5	-	-
16	32	-	-	16	Sulphur
3	-	4	3	-	-
-	16	-	8	8	Oxygen

(4 marks-[An/C],2)

B2 Figure 7- a, b and c illustrates various classification of Periodic table at different time line



Type	Elements	Atomic Mass
A	Li Na K	7 23 39
B	N P As	14 31 74
C	B Al Ga	10.8 27 69.7

Figure 7 Evolution of periodic table (a). John Newland's; (b). Dobereine's; (c). Dmitri Mendeleev; (d). Modern Periodic Table

(i) From the four periodic classifications mentioned in the figure 7, identify the classification where all the elements in a given group have the same valence electrons. Justify your answer by indicating the electronic configuration of two elements in a group.

(3 marks-[U/C], 2)

(ii) Categorize the following elements that obeys and fails Dobereine's triads. [Figure 7b]. Choose the reason for failure from options (A-D).

a. $_{20}Ca^{40}$, $_{53}Sr^{88}$, $_{56}Ba^{137}$
b. $_{17}Cl^{35.5}$, $_{35}Br^{80}$, $_{53}I^{127}$,
c. $_{26}Fe^{56}$, $_{27}Co^{59}$, $_{28}Ni^{59}$

Option A: Due to same/nearest atomic mass
Option B: Due to increase in number of neutrons
Option C: Due to change in the valence shell orbital and electrons
Option D: Existence of equal ratio of isotope of element

(3 marks-[Ap/C], 2)

(iii) Understand the John Newland's classification [Figure 7a] and answer the following questions

a. From figure 7a, identify the element pairs which possess similar properties.
b. Verify whether the elements in table 4 are arranged based on Newland's classification (Yes/No). Validate your answer.

Table 4

Atomic No	13	14	15	16	17	18	19	20	21
Elements	Al	Si	P	S	Cl	Ar	K	Ca	Sc

(4 marks-[U/C], 2)

B3	In the modern periodic table a correlation on elemental properties are interconnected. For example: Electronegativity is the function of ionization energy.	

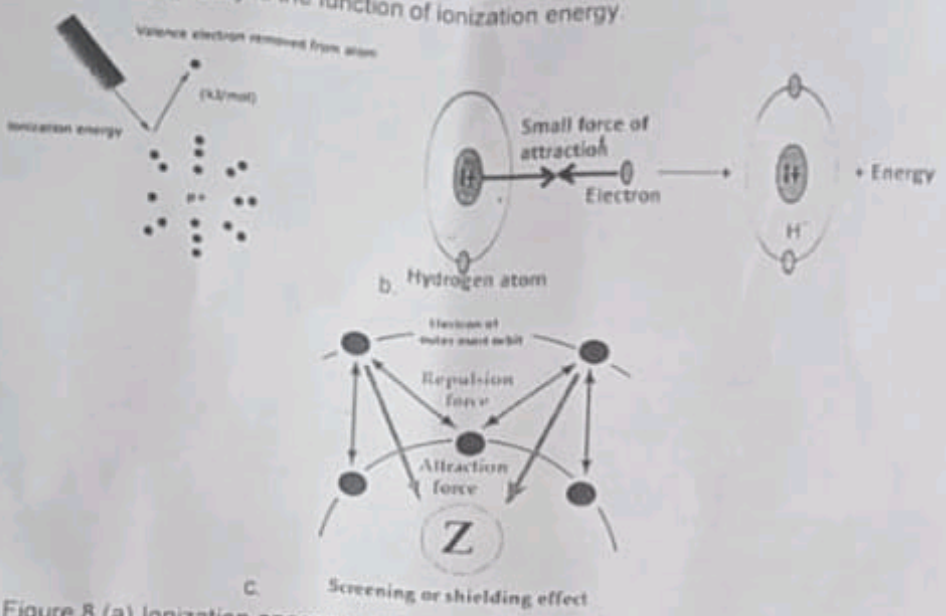


Figure 8 (a) Ionization energy, (b) Electron affinity, (c) Effective nuclear charge

(i)	Ionization energy of Boron [$_5B^{10}$] is less than Beryllium [$_4Be^8$] even though the nuclear charge of Boron is higher. Predict the reason. [Refer Fig 8a]	(3 marks- [Ap/C], 3)
(ii)	The atomic number of Fluorine is nine. a. Write the electronic configuration/arrangements in various orbital for fluorine atom (F) and its anion (F^-). b. Assess the difference in the electron affinity of fluorine atom and its anion [Refer Fig 8b]	(3 marks-[Ap/C], 2)
(iii)	Following are the electronic configuration of the four atoms in the periodic table. Based on the scenario answer the following question. Atom A: $1s^2 2s^2 2p^5$ Atom B: $1s^2 2s^2 2p^4$ Atom C: $1s^2 2s^2 2p^3$ Atom D: $1s^2 2s^2 2p^2$ a. Arrange the ascending order of effective nuclear charge for the given atoms (A, B, C and D). b. Atom A and D has higher and lower electronegativity respectively. Based on scenario, predict the reason. [Refer Fig 8c]	(4 marks-[Ap/C], 2)

SECTION C	COURSE OUTCOME 3	MAXIMUM: 10 MARKS

C1	The Lewis dot structure represent the outermost electron in an atom whereas the octet rule prefer to have eight electrons in the valence shell. Critically view the figure and answer the following questions.

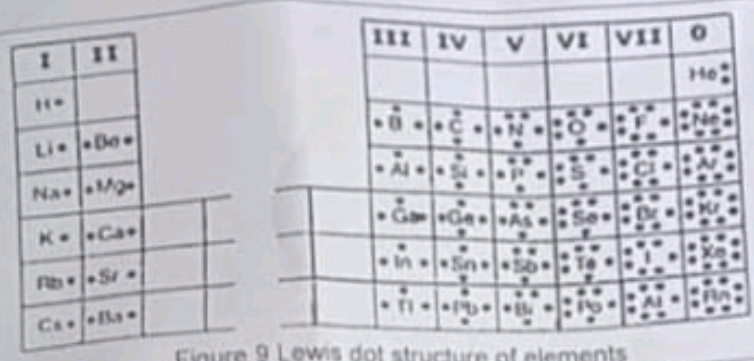


Figure 9 Lewis dot structure of elements

(i) Statement: Aluminium chloride ($AlCl_3$) is unstable based on octet rule.
Reason: Aluminium in the compound $AlCl_3$ possess eight electrons in the outermost shell.
Select the right option from the following. Give reason for the answer.
(a) Statement is right but reason is wrong
(b) Statement is wrong but reason is right
(c) Both the statement and reason are right
(d) Both the statement and reason are wrong

(2 Marks – [Ap/C],2)

(ii) Draw the Lewis structure of the following compounds and predict if the compound(s) obey the octet rule. Give reason for the answer. [Refer scenario]
(a) BF_3 (b) NaCl

(3 Marks – [Ap/C],2)

C2 Sodium fluoride (NaF) is used in toothpaste (refer fig. 10). It is an inorganic compound soluble in water and has a crystal structure similar to NaCl.

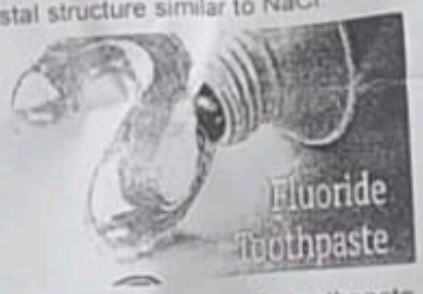


Figure 10 Sodium fluoride toothpaste

(i) Sodium fluoride is soluble in water but insoluble in n-hexane. Reason out. [Refer scenario]

(2 Marks – [An/C],2)

(ii) The electronic configuration of Na and F are shown below.
Sodium = $1s^2\ 2s^2\ 2p^6\ 3s^1$
Fluorine = $1s^2\ 2s^2\ 2p^5$

Sodium combine with fluorine to form sodium fluoride. Identify the type of bond formation and predict the number of outermost valence electrons present in sodium (Na^+) and fluoride (F^-) ions.

(3 Marks – [An/C],2)

C3 The nail polish remover contain acetone (CH_3-CO-CH_3), linked via covalent bond. The electronegativity of carbon, hydrogen and oxygen are 2.5, 2.1, and 3.5 respectively.

Figure 11 Molecular Formula of nail polish remover

(i) In acetone, carbon and oxygen linked by double bonds. Predict the reason for the formation of double bond between them. [Refer figure 11]

(2 Marks – [Ap/C],2)

(ii) Acetone comprises three different bonds as listed in the table. [Refer figure 11] Identify the type of bond as either polar or non-polar covalent bond (based on electronegativity) and indicate their charge separation given in table 5.

Table 5

S.No	Bonds present in acetone	Polar/Non-polar covalent	δ+ / δ- (eg., $H^{\delta+}/OH^{\delta-}$)
1	C-H		
2	C-C		
3	C-O		

(3 Marks – [U/C],2)

End of Question Paper